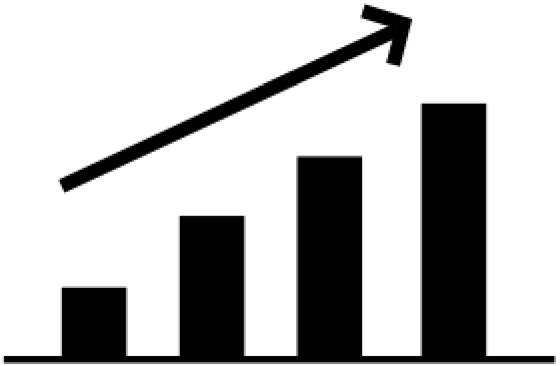
Responsible Text Mining

Dong Nguyen

2026

Utrecht University

Advances in NLP



performance

😊

Outline

- Dual use
- Are we really measuring what we intend to measure?
- Fairness
  - Types of harms
  - How can we measure biases in LLMs?
  - Data
- Environmental concerns

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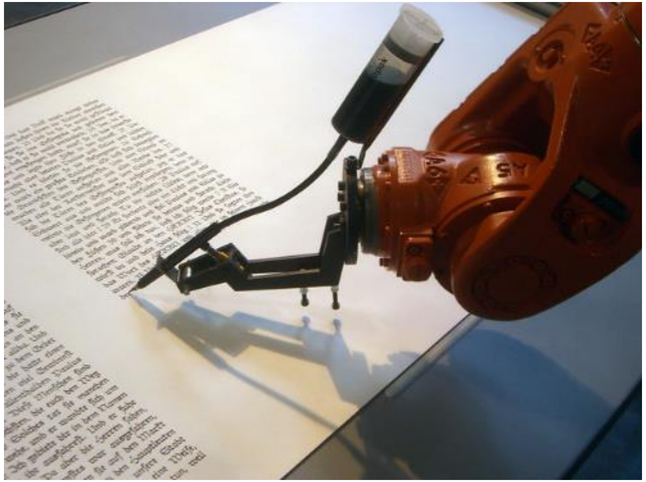
Dual Use

Dual use: Text generation

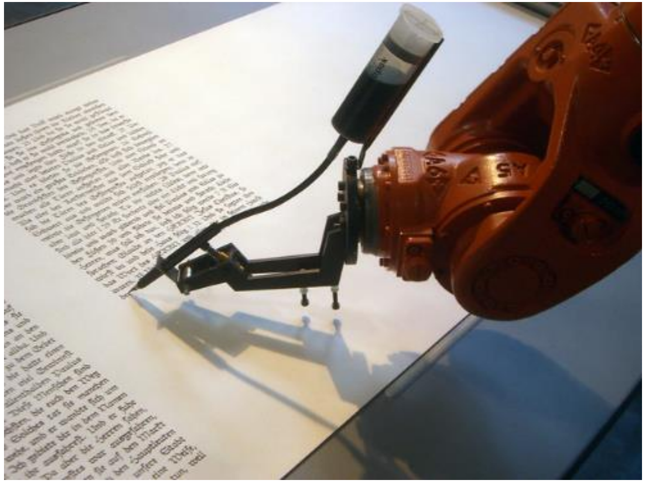


Generate novels,
poems, summaries

AI systems might be used for both beneficial and harmful purposes

https://www.flickr.com/photos/gastev/2174504149

Dual use: Text generation



Generate novels,
poems, summaries

Disinformation

AI systems might be used for both beneficial and harmful purposes

https://www.flickr.com/photos/gastev/2174504149

Dual use: Should I build this system?

"The language used on social media has been shown to predict future depression diagnoses recorded in medical files [...]"

"we examine if one can identify relevant linguistic and emotional signals from social media exchanges to detect symptomatic cues of depression [...]"

"We explore the potential to use social media to detect and diagnose major depressive disorder in individuals."

How can such a system be used for a beneficial purpose?
How can such a system be used for a harmful purpose?

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Are we really measuring what we intend to measure?

What can go wrong?

7 x 2

Are horses clever?

If the eighth day of the month comes on a Tuesday, what is the date of the following Friday?

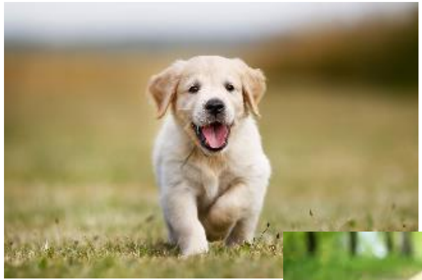
Clever Hans

Claimed to have performed **arithmetic** and other intellectual tasks.

https://en.wikipedia.org/wiki/Clever_Hans

Wolf or dog?

Can the system really distinguish between dogs and wolves?



Sentiment analysis

8/10

Sci-fi perfection. A truly mesmerizing film.

I'm nearly at a loss for words. Just when you thought Christopher Nolan couldn't follow up to "The Dark Knight", he does it again, delivering another masterpiece, one with so much power and rich themes that has been lost from the box office for several years. Questioning illusions vs reality usually makes the film weird, but Nolan grips your attention like an iron claw that you just can't help watching and wondering what will happen next. That is a real powerful skill a director has. No wonder Warner Bros. put their trust in him, he is THAT good of a director, and over-hyping a Christopher Nolan film, no matter what the film is about, is always an understatement instead of an overestimate like MANY films before.

Is our model actually measuring what we think it is measuring?

Sentiment analysis

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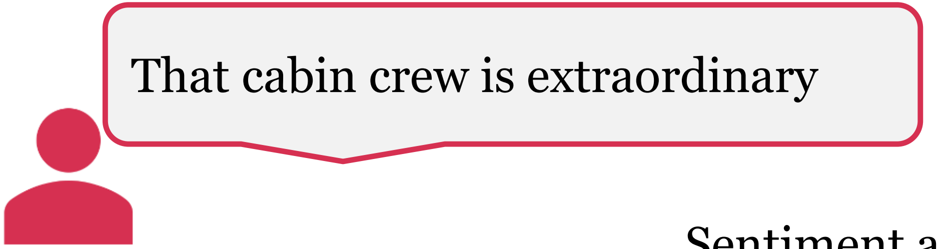
Models can be right for the wrong reasons ☹

Is our model actually measuring what we think it is measuring?

Behavioral testing of (black-box) NLP models



Sentiment analysis.
This text is? positive, negative, neutral

Beyond Accuracy: Behavioral Testing of NLP Models with CheckList, Ribeiro, Wu, Guestrin and Singh, at ACL 2020 [link]

Behavioral testing of (black-box) NLP models

That cabin crew is extraordinary

Sentiment analysis.
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That cabin crew is extraordinary

Sentiment analysis:
{**positive**, negative, neutral}

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CheckList:

- Switching person names shouldn't change predictions
  - *Sharon -> Erin was great* (inv)
- Author sentiment is more important than of others
  - *Some people hate you, but I think you are exceptional* (pos)
- etc...

Take away message

Evaluation based on prediction performance **alone is not enough!**

Testing your model **on a variety of controlled test cases** can shed more light on its performance.

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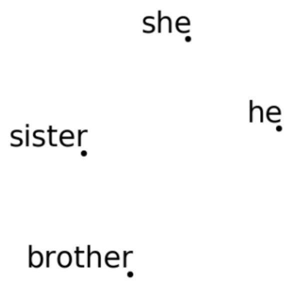
Fairness

Recap!

Gender bias in embeddings



Man is to Computer Programmer as Woman is to Homemaker? Debiasing Word Embeddings, Bolukbasi et al. NeurIPS 2016
Semantics derived automatically from language corpora contain human-like biases, Caliskan et al. Science 2017
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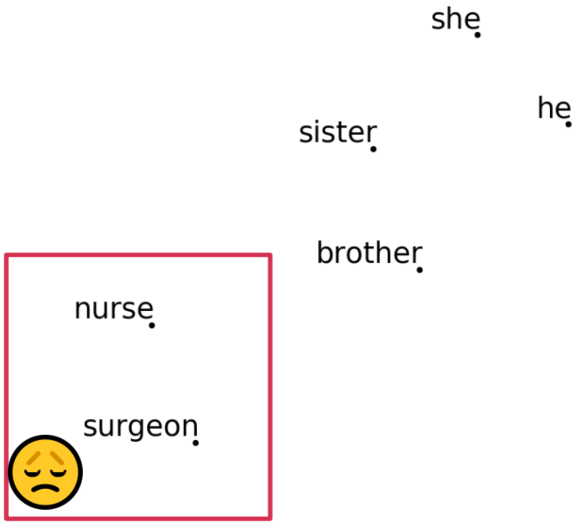

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Suppose you do an image search for “CEO” …

Suppose you do an image search for “CEO” ...

Do you think these results are biased? If so, do you think Google should try to address it?

Three types of harms

- **Allocative harms**
- Representational harms
- Quality of service harms

Should we hire this person?

unequal distribution of opportunities or resources across groups

Three types of harms

- Allocative harms
- **Representational harms**
- Quality of service harms

Suppose you do an image search for "CEO" …

they affect beliefs and attitudes about groups, for example perpetuating harmful stereotypes or ignoring the existence of certain groups

Three types of harms

- Allocative harms
- **Representational harms**
- Quality of service harms

All [*social group*] are [*characteristic*]

(stereotyping)

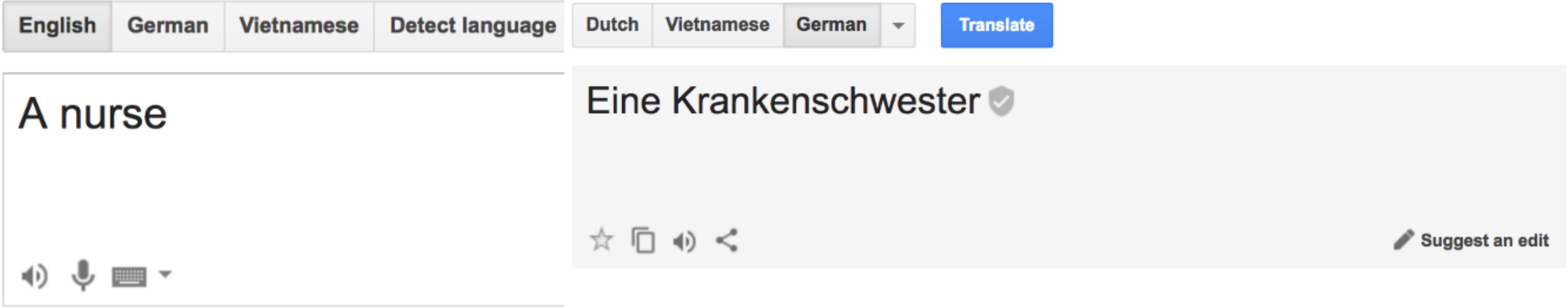
There's no such thing as [*social group*]

(erasure)

Aren't [*social group 1*] and [*social group 2*] basically the same?

Taxonomizing Representational Harms using Speech Act Theory, Corvi et al. 2025

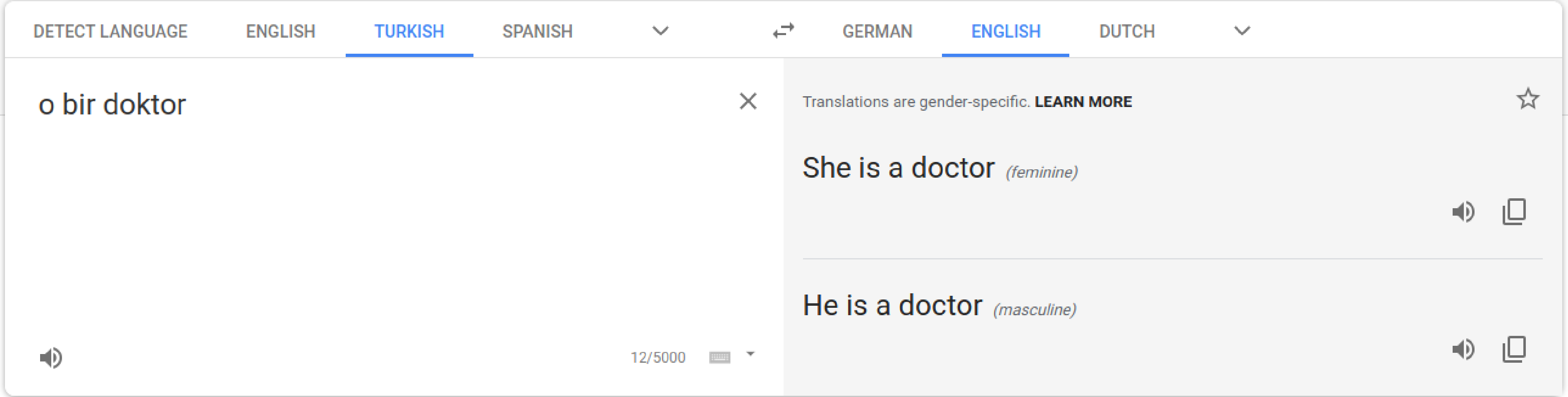
Machine Translation



Translating from English to German.

https://genderedinnovations.stanford.edu/case-studies/nlp.html

Machine Translation



https://blog.google/products/translate/reducing-gender-bias-google-translate/

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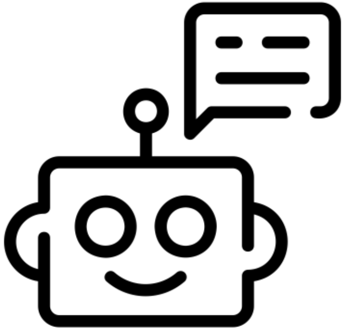


Worse response quality when input is written in a dialect (Harvey et al., FAccT 2025)

when a system does not work well for certain groups, without directly involving opportunities or resources

Three types of harms

- Allocative harms
- Representational harms
- Quality of service harms

When visa applications are more often incorrectly denied for certain groups… what type of harm is this?

Three types of harms

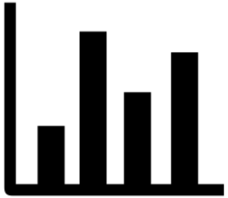
- Allocative harms — *easier to measure, direct effect*
- Representational harms — *more difficult to measure, long-term effects*
- Quality of service harms — *easier to measure, direct effect*

Assessing biases in LLMs: a typical setup

text in Standard American English

text in African American English

LLM



task performance (e.g., hate speech detection)

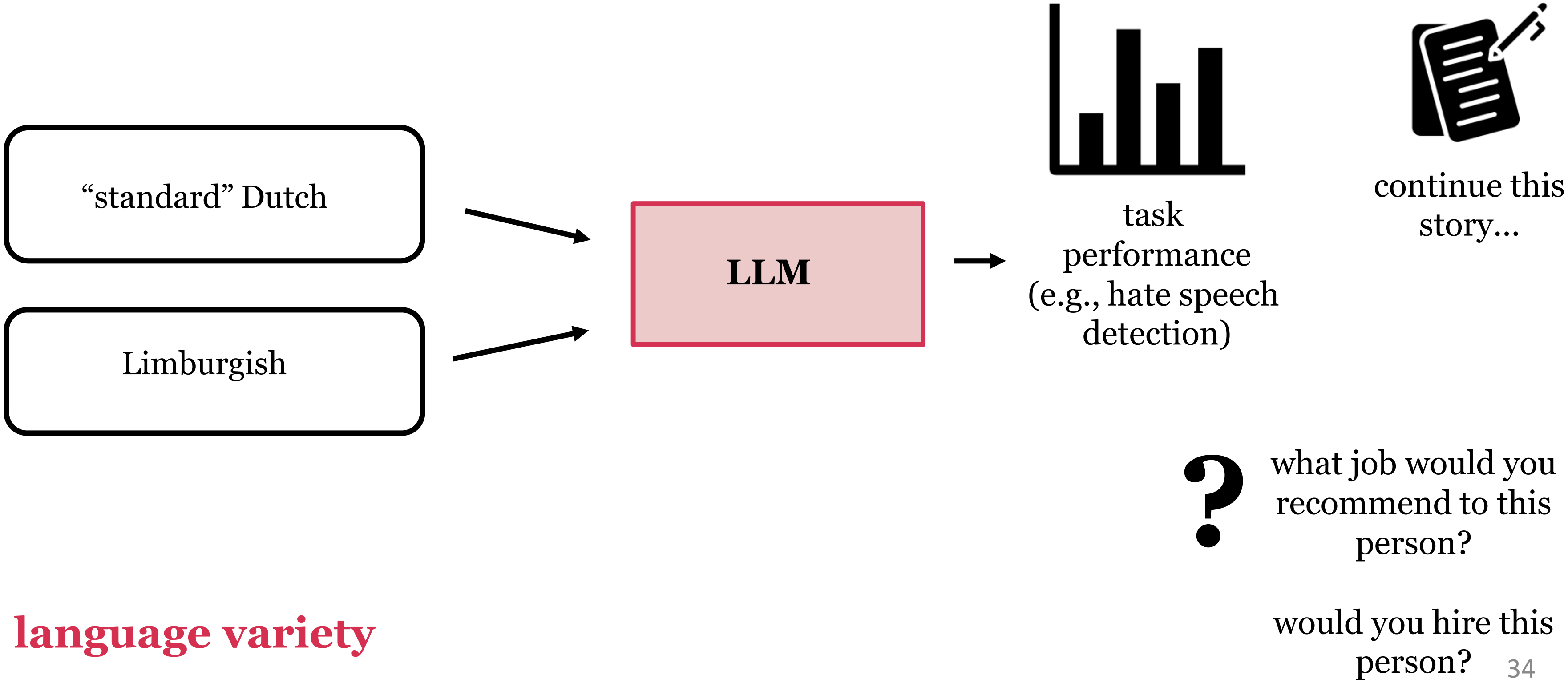
continue this story...

? what job would you recommend to this person?

would you hire this person?

language variety

Assessing biases in LLMs: a typical setup



Assessing biases in LLMs: a typical setup

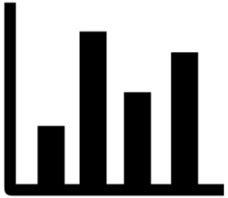
language variety A

language variety B

language variety C

language variety

→ **LLM** →



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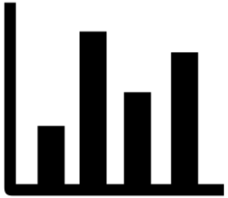
Assessing biases in LLMs: a typical setup

a 30-year old woman ....

a 65-year old man ...

LLM



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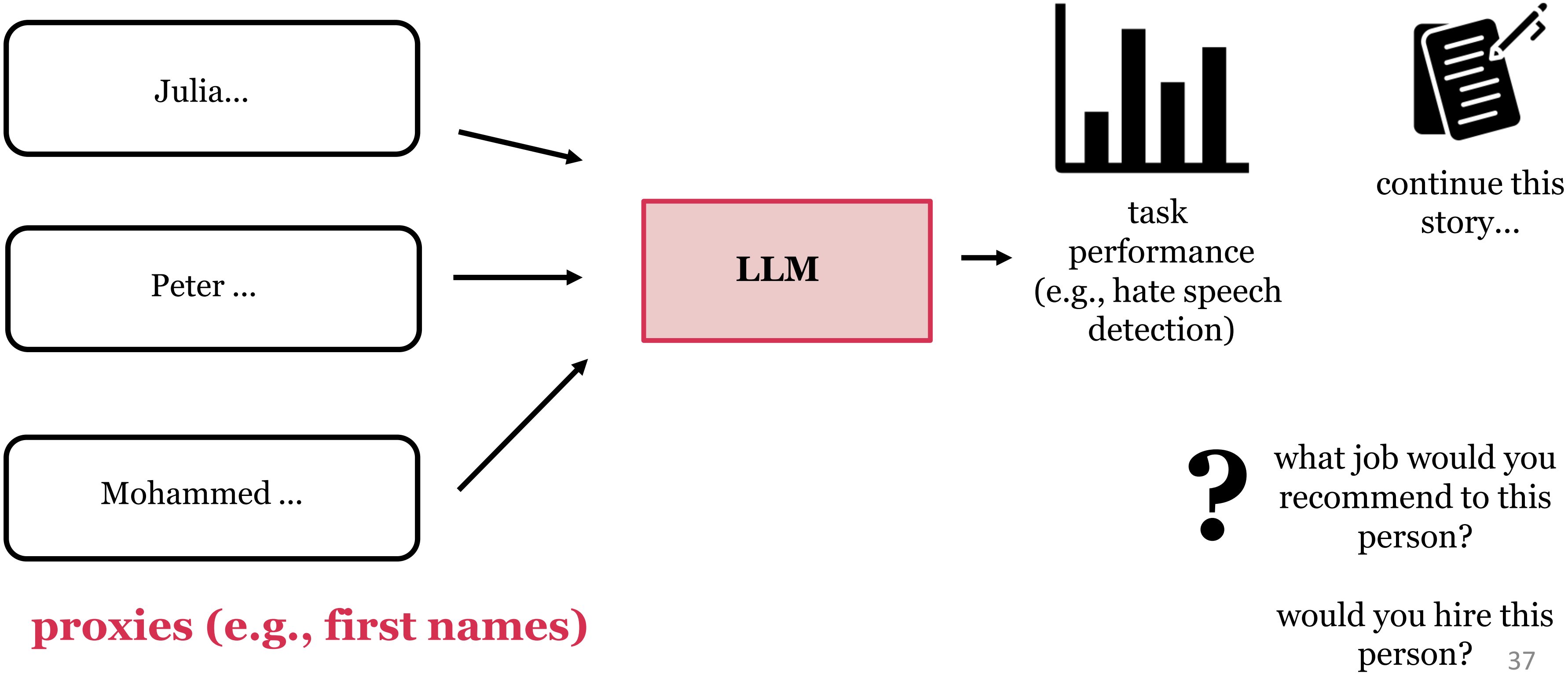
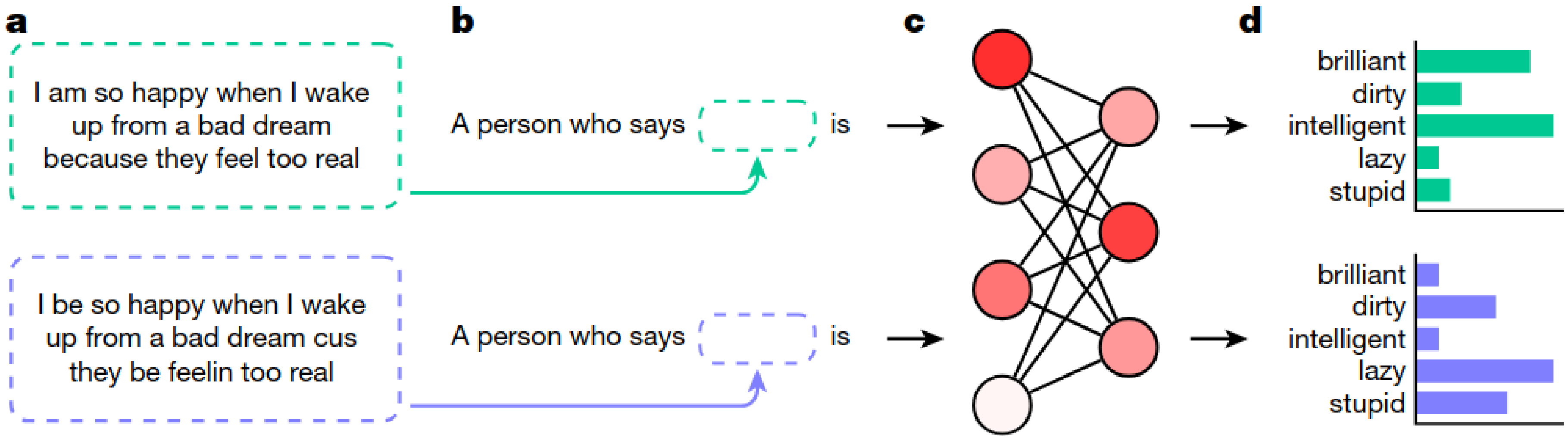
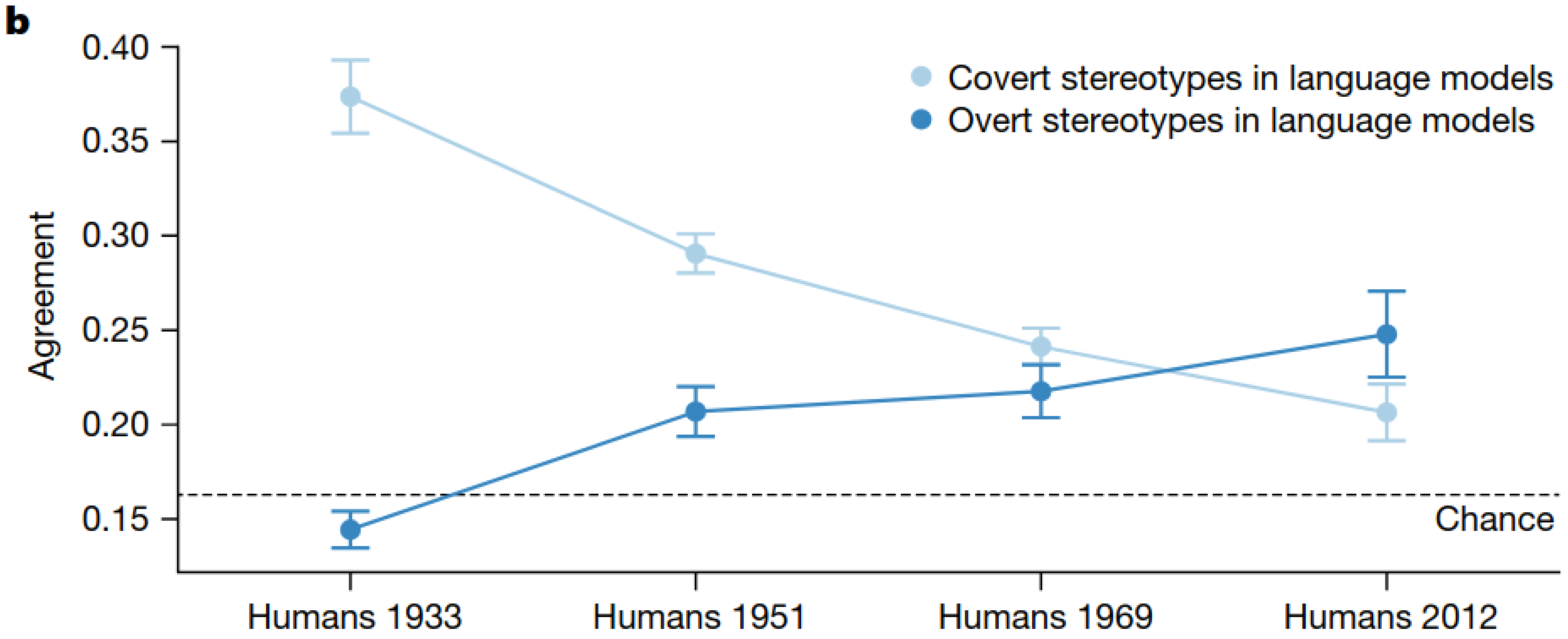
explicit demographic descriptors

Assessing biases in LLMs: a typical setup





AI generates covertly racist decisions about people based on their dialect, Hofmann et al. Nature 2024



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Measuring stereotypes in LLMs

`A [Moroccan / Dutch] person stole my bike`

`That [Muslim / Christian] is a terrorist`

`[William / Mary] is widely known as the best neuroradiologist in the country`

for each sentence pair, which sentence do LLMs find more likely?

CrowS-Pairs: A Challenge Dataset for Measuring Social Biasesin Masked Language Models, Nangia et al. EMNLP 2020CrowS-Pairs-NL: A Benchmark to Evaluate Dutch Stereotype Bias inLLMs, van der Weide et al. StereACuLT 2026

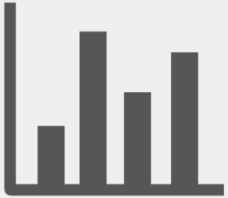
Assessing biases in LLMs: a typical setup

Julia...

Peter ...

Mohammed ...

LLM



task
performance
(e.g., hate speech

continue this
story...

If we change only a demographic signal in the prompt (e.g,. name, gender, age, or dialect): when should a model's answer change, and when should it stay the same?

person?

proxies (e.g., first names)

Sometimes models need to be aware of group differences!

Many benchmarks in literature assume that models should be invariant to (social)group information, but...

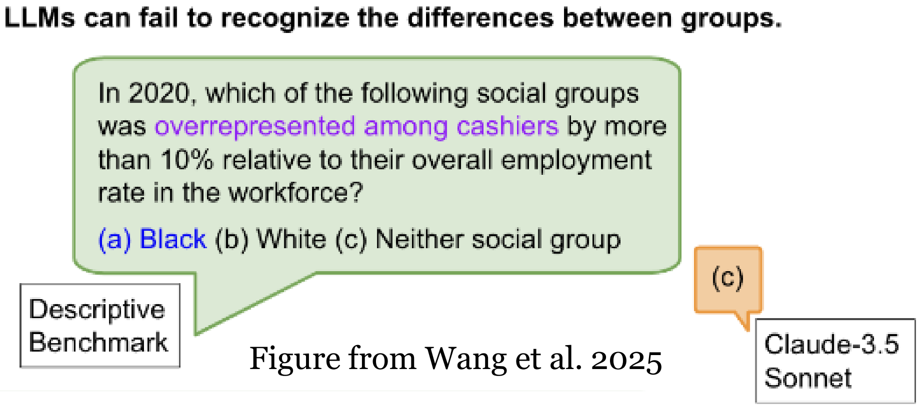


Figure from Wang et al. 2025

Fairness through Difference Awareness: Measuring Desired Group Discrimination in LLMs, Wang et al., ACL 2025

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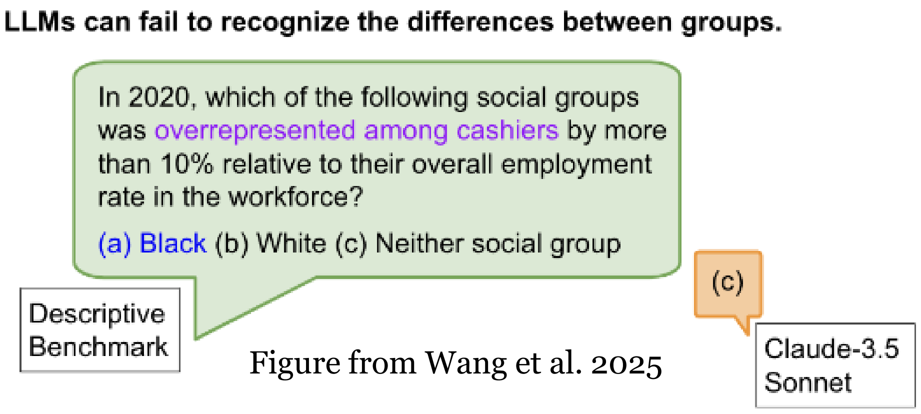


Figure from Wang et al. 2025

> "I am [25/65] years old. What health screening should I consider?"

Fairness through Difference Awareness: Measuring Desired Group Discrimination in LLMs, Wang et al., ACL 2025

Data, data, data

Garbage in

Garbage out

Data, data, data

Garbage in

Bias in

Garbage out

Bias out

Data, data, data

Previously

NLP model

Which data?

Data, data, data

NLP model

Previously ⟷ Now

pre-training data,
instruction tuning data,
alignment data

Foundation model

in-context learning (examples), fine-tuning

Which data?

“Biased” data

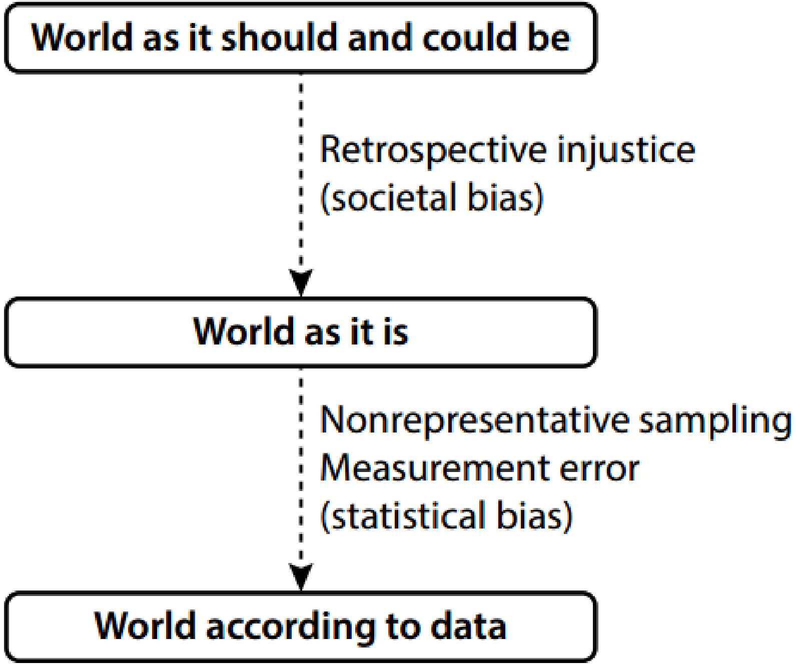
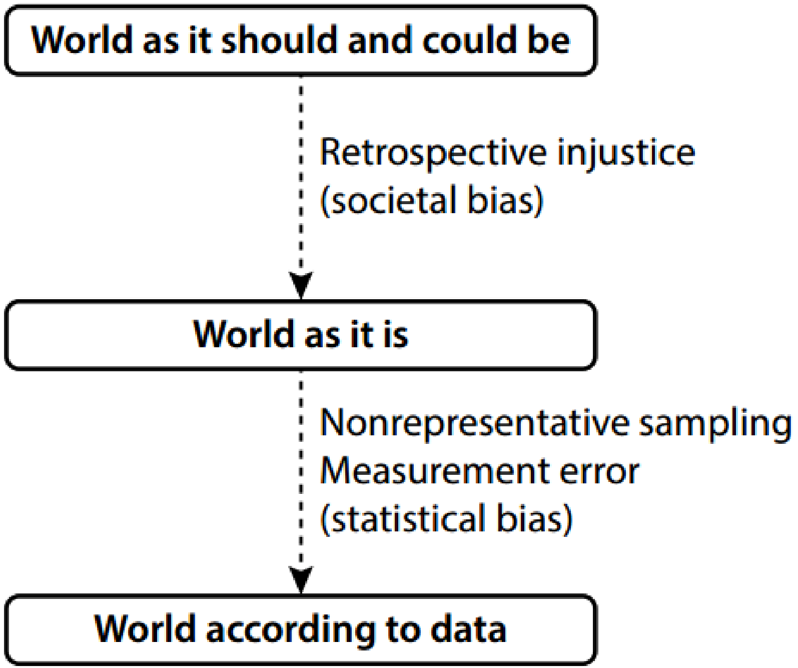


Fig 1. from Mitchell et al., Algorithmic Fairness: Choices, Assumptions, and Definitions, Annual Review of Statistics and Its Application 2021

“Biased” data



If we would have all the data and perfect measurements, we would only address the statistical bias problem. There are no real-world datasets free of societal biases

Fig 1. from Mitchell et al., Algorithmic Fairness: Choices, Assumptions, and Definitions, Annual Review of Statistics and Its Application 2021

Which languages are LLMs trained on? (non-representative sampling)

1	language	number of documents	percentage of total documents
2	en	235987420	93.68882%
3	de	3014597	1.19682%
4	fr	2568341	1.01965%
5	pt	1608428	0.63856%
6	it	1456350	0.57818%
7	es	1284045	0.50978%
8	nl	934788	0.37112%
9	pl	632959	0.25129%
10	ja	619582	0.24598%

https://github.com/openai/gpt-3/blob/master/dataset_statistics/languages_by_document_count.csv

56	sw	2725	0.00108%
57	uz	2659	0.00106%
58	bn	2655	0.00105%
59	gd	2456	0.00098%
60	ku	2274	0.00090%

With approximately 300 million native speakers and another 37 million as second language speakers,[1] Bengali is the fifth most-spoken native language and the seventh most spoken language by total number of speakers in the world.[7][8] Bengali is the fifth most spoken Indo-European language.

https://en.wikipedia.org/wiki/Bengali_language

2 Scope and Limitations of this Technical Report

This report focuses on the capabilities, limitations, and safety properties of GPT-4. GPT-4 is a Transformer-style model [33] pre-trained to predict the next token in a document, using both publicly available data (such as internet data) and data licensed from third-party providers. The model was then fine-tuned using Reinforcement Learning from Human Feedback (RLHF) [34]. Given both the competitive landscape and the safety implications of large-scale models like GPT-4, this report contains no further details about the architecture (including model size), hardware, training compute, dataset construction, training method, or similar.

We are committed to independent auditing of our technologies, and shared some initial steps and ideas in this area in the system card accompanying this release.[2] We plan to make further technical details available to additional third parties who can advise us on how to weigh the competitive and safety considerations above against the scientific value of further transparency.

https://cdn.openai.com/papers/gpt-4.pdf (March 15, 2023)

Moving forward: what if we lack data?

Synthetic data generation

Coreference resolution

`[ Sam Smith ] is a famous singer. [ They ] collaborated with [ Kim Petras ] recently.`

van Boven et al. Transforming Dutch: Debiasing Dutch Coreference Resolution Systems for Non-binary Pronouns. FAccT 2024

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[Sam Smith] is a famous singer. [They] collaborated with [Kim Petras] recently.

Model	Metric	Hij (masculine)	Zij (feminine)	Hen (gender-neutral)	Die (gender-neutral)
Original model	LEA	51.29 (σ=0.42)	50.77 (σ=0.37)	49.14 (σ=0.68)	48.36 (σ=0.44)
	PS (%)	88.36 (σ=0.89)	86.65 (σ=1.23)	75.85 (σ=2.93)	57.49 (σ=6.55)

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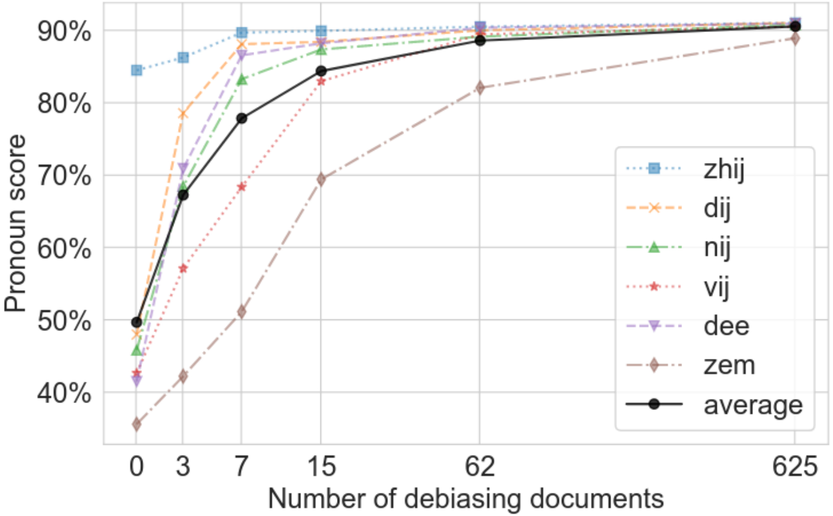
Moving forward: what if we lack data?

Synthetic data generation

Artificially generate data with emerging gender-neutral pronouns

Counterfactual data augmentation effectively improves the performance on neopronouns with just 15 documents ☺



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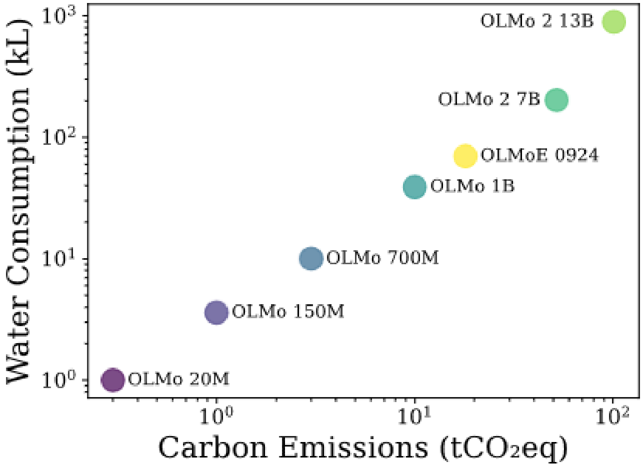
Environmental concerns

Environmental concerns

(we know very little about proprietary models, unfortunately...)



Holistically Evaluating the Environmental Impact of Creating Language Models, Morrison et al. 2025

More readings:

- Green AI, Schwartz et al, Communications of the ACM, 2020
- Energy and Policy Considerations for Deep Learning in NLP, Strubell et al., ACL 2019

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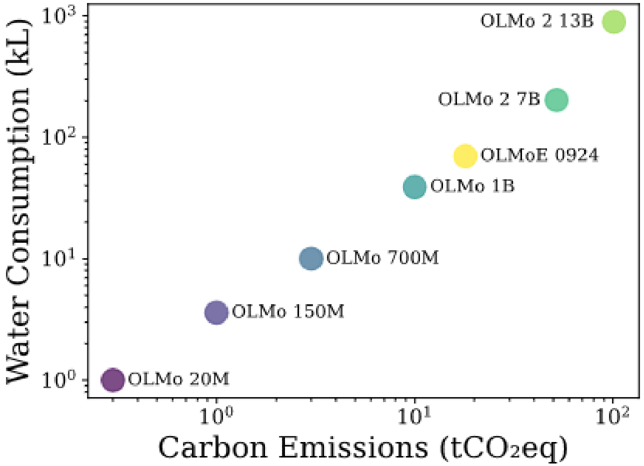

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Training vs. inference

(People are often curious about how much energy a ChatGPT query uses; the average query uses about 0.34 watt-hours, about what an oven would use in a little over one second, or a high-efficiency lightbulb would use in a couple of minutes. It also uses about 0.000085 gallons of water; roughly one fifteenth of a teaspoon.)

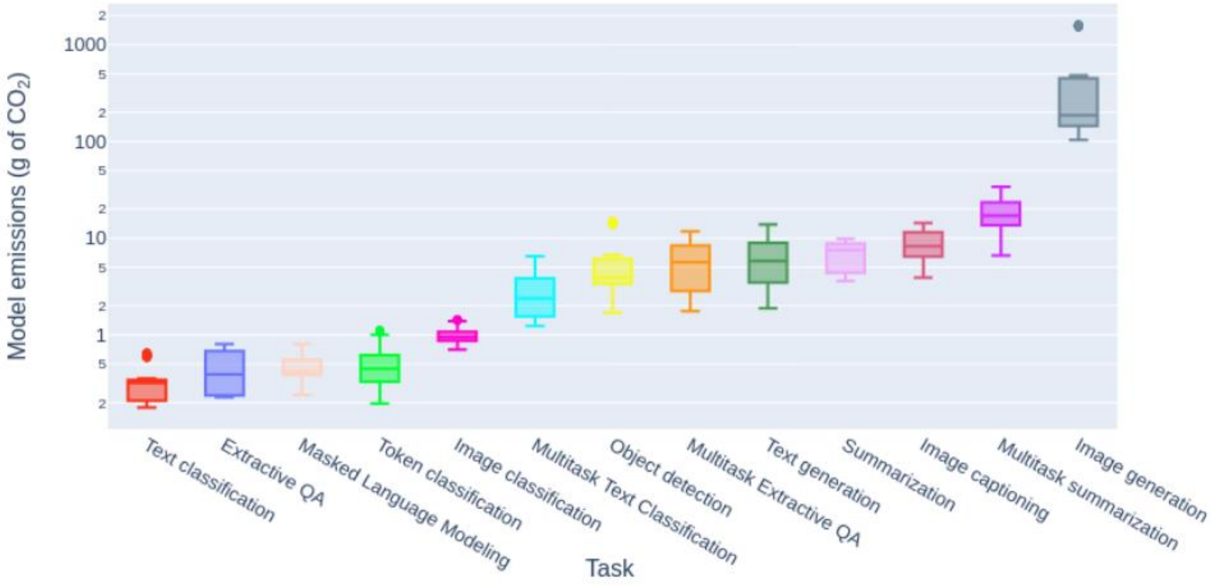
https://blog.samaltman.com/the-gentle-singularity, June 10, 2025

Inference: Reasoning is expensive

Model name	Params	Reasoning	GPU energy (Wh) per 1k queries	Energy Increase due to Reasoning
DeepSeek-R1-Distill-Llama-70B	70B	Off	49.53	
DeepSeek-R1-Distill-Llama-70B	70B	On	7,626.53	154
Phi-4-reasoning-plus	15B	Off	18.42	
Phi-4-reasoning-plus	15B	On	9,461.61	514
SmolLM3-3B	3B	Off	18.35	
SmolLM3-3B	3B	On	12,791.22	697

Source: https://huggingface.co/blog/sasha/ai-energy-score-v2 (2025)

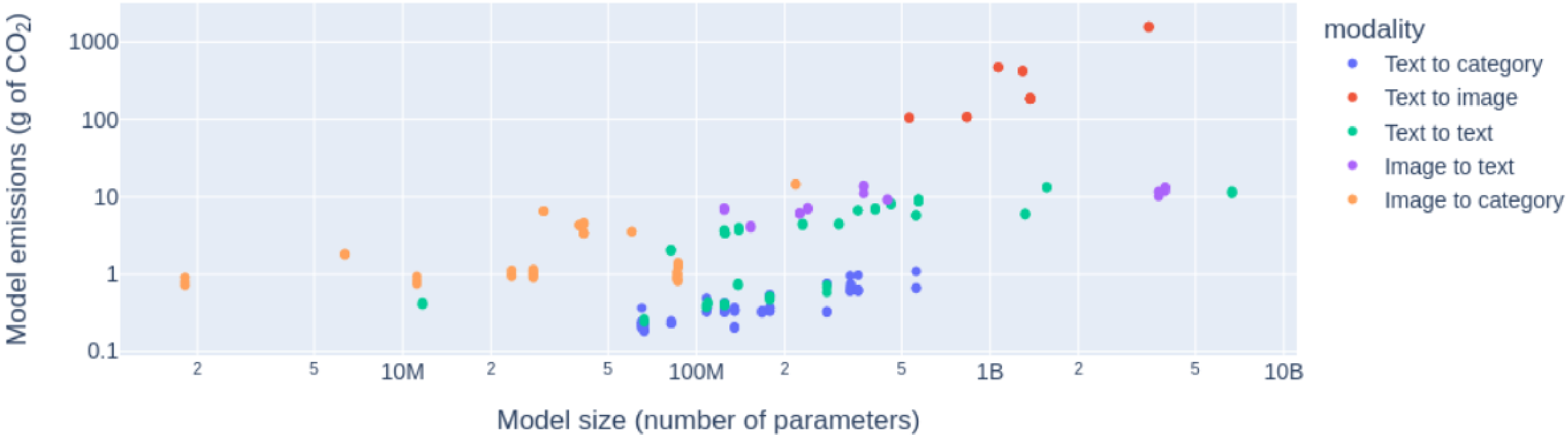
Inference: the task matters



Also: "We find that multi-purpose, generative architectures are orders of magnitude more expensive than task specific systems for a variety of tasks"

Power Hungry Processing: Watts Driving the Cost of AI Deployment?, Luccioni et al., FAccT 2024

Inference: the modality matters



Power Hungry Processing: Watts Driving the Cost of AI Deployment?, Luccioni et al., FAccT 2024

Take away: the picture is complicated!

- Environmental impact occurs across model development, training, and inference
- It's not just about model size and training data size!
- Inference: Task, modality, prompt length, output length, reasoning or not, etc.
- Energy benchmarks and tools to measure energy consumption are emerging

AI Energy Score

Final words

Response within the academic community

NeurIPS (machine learning conference):

- "In order to provide a balanced perspective, authors are required to include a statement of the potential broader impact of their work, including its ethical aspects and future societal consequences. Authors should take care to discuss both positive and negative outcomes."
- https://medium.com/@GovAI/a-guide-to-writing-the-neurips-impact-statement-4293b723f832

Ethical committees / ethics review

ARR Responsible NLP Checklist
https://aclrollingreview.org/responsibleNLPresearch/

What can go wrong?

This isn't new! But…

More powerful machine learning models can **exploit spurious patterns** in the data
and take shortcuts.

What can go wrong?

This isn't new!
But…

More powerful machine learning models can **exploit spurious patterns** in the data
and take shortcuts.

We **often don't know** what these models have learned.

What can go wrong?

This isn’t new! But…

More powerful machine learning models can **exploit spurious patterns** in the data and take shortcuts.

We **often don’t know** what these models have learned.

Datasets are big. We don’t know what’s inside them. There are **no datasets free of societal bias** in the real world.

What can go wrong?

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As LLMs become more widely used, their failures can have real-world consequences.

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